Mohammad Elayan, MSc

Graduate Research & Teaching Assistant

Dept. of Civil & Environmental Engineering
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Profile

PhD student and Graduate Research Assistant at the University of Nebraska-Lincoln with a decade of professional experience in transportation engineering in Dubai, UAE, contributing to 100+ projects across the UAE and MENA region. Research spans traffic agent heterogeneity, automated vehicle behavior, pedestrian safety, and multi-modal transport modeling. Co-authored 10 publications and delivered 8 conference presentations.

Education

#	Degree — Program & Institution	Grade	Year
1	**PhD** — Civil Engineering (Transportation Engineering) *University of Nebraska-Lincoln, Lincoln, NE, USA*	Ongoing	2023–Present
2	**MSc** — Civil Engineering (Transportation Engineering) *Jordan University of Science & Technology, Irbid, Jordan*	91.6% · Rank 1	2011–2013
3	**BSc** — Civil Engineering (Transportation Engineering) *Jordan University of Science & Technology, Irbid, Jordan*	82.3% · Rank 16	2006–2011

Research

Peer-Reviewed Publications (7)

1. **Elayan, M.**, & Kontar, W. (2026). "Can Automated Vehicles Have it All? A Consensus Framework for Diagnosing Behavioral Trade-Offs in Mixed Traffic." *Transportation Research Part C: Emerging Technologies* (Accepted).
2. **Elayan M.**, & Kontar, W. (2026). "Consensus-Aware AV Behavior: Trade-Offs Between Safety, Interaction, and Performance in Mixed Urban Traffic." *2025 IEEE 28th International Conference on Intelligent Transportation Systems (ITSC)*, Gold Coast, Australia. [Link]
3. **Elayan, M.**, Karki, S., & Hawkins, J. (2025). "Better Safety Analyses through Smarter Data: Adding Open-Street-View and Traffic Calibrated-LBS Data to Pedestrian Crash Analysis in Lincoln, NE." *Transportation Research Record*. [Link]
4. **Elayan, M.**, Aldridge, N., Hawkins, J., & Nam, Y. (2025). "Integrating StreetLight, EPS Smart Location Data and Road Attributes: A Random Forest Approach to Multi-Modal Traffic Calibration in Lincoln, Nebraska." *Journal of Transportation Engineering, Part A: Systems*, Vol. 151(8). [Link]
5. Al-Khateeb, G., Al-Suleiman, T., Khedaywi, T., & **Elayan, M.** (2016). "Studying Rutting Performance of Superpave Mixtures." *Road Materials and Pavement Design*, Vol. 19(2). [Link]
6. Alsheyab, M., Khedaywi, T., & **Elayan, M.** (2013). "Laboratory Study on Solidification/Stabilization of Unwanted Medications Using Asphalt as a Binder." *Journal of Material Cycles and Waste Management*, Vol. 15(2). [Link]
7. Al-Suleiman, T., & **Elayan, M.** (2013). "Gap Acceptance Behavior at U-turn Median Openings – Case Study in Jordan." *Jordan Journal of Civil Engineering*, Vol. 7(3). [Link]

Other Publications (3)

1. **Elayan, M.**, & Kontar, W. (2026). “Learning the Pareto Space of Multi-Objective Autonomous Driving: A Modular, Data-Driven Approach.” *arXiv preprint arXiv:2601.18913*. [Link]
2. **Elayan, M.**, & Kontar, W. (2025). “Consensus-Aware AV Behavior: Trade-offs Between Safety, Interaction, and Performance in Mixed Urban Traffic.” *arXiv preprint arXiv:2505.04379*. [Link]
3. Nam, Y., Hawkins, J., Butler, D., Aldridge, N., **Elayan, M.**, & Yoo, J. (2024). “Modeling Pedestrian and Bicyclist Crash Exposure with Location-Based Service Data.” *Nebraska Dept. of Transportation, Technical Report No. SPR-FY23(025)*. [Link]

Conference Papers & Presentations (8)

1. **Elayan, M.**, & Kontar, W. (2026). “Behavioral Heterogeneity as Quantum-Inspired Representation.” *IEEE ITSC*, Naples, Italy. [Accepted]
2. **Elayan, M.**, & Kontar, W. (2026). “Learning the Pareto Space of Multi-Objective Autonomous Driving: A Modular, Data-Driven Approach.” *IEEE Intelligent Vehicles Symposium (IV)*, Detroit, MI. [Accepted]
3. **Elayan, M.**, & Kontar, W. (2026). “Consensus-Aware AV Behavior: Empirical Pareto-Guided Reinforcement Learning for Mixed Traffic.” *NSF NAIRR Annual Meeting*, Arlington, VA. [Presented]
4. **Elayan, M.**, & Kontar, W. (2026). “The Empirical Pareto Frontier of Automated Driving.” *College of Engineering 7th Annual Graduate Student Symposium*, UNL. [Presented]
5. **Elayan, M.**, & Kontar, W. (2026). “A Unified Framework for Consensus-Aware AV Behavior.” *TRB 105th Annual Meeting*, Washington, DC. [Presented]
6. Armantalab, O., **Elayan, M.**, & Hawkins, J. (2025). “Multi-Stage Clustering of Daily Activity Time Series.” *TRB 105th Annual Meeting*, Washington, DC. [Presented]
7. **Elayan, M.**, & Kontar, W. (2025). “Consensus-Aware AV Behavior: Trade-offs Between Safety, Interaction, and Performance in Mixed Urban Traffic.” *IEEE ITSC*, Gold Coast, Australia. [Presented]
8. **Elayan, M.**, Karki, S., & Hawkins, J. (2025). “Better Safety Analyses through Smarter Data.” *TRB 104th Annual Meeting*, Washington, DC. [Presented]

Under Review (1)

1. Armantalab, O., **Elayan, M.**, & Hawkins, J. (2026). “Multi-Stage Clustering of Daily Activity Time Series: Patterns and Socio-Demographic Insights.” *Transportmetrica A: Transport Science*.

Research Projects

- *Modeling Pedestrian and Bicyclist Crash Exposure with Location-Based Service Data.*
  Nebraska Department of Transportation, USA. 2022–2024
- *Introducing New Technologies and Advanced Methodologies for Asphalt Mixtures of Highway Pavements in Jordan.*
  Ministry of Higher Education and Scientific Research, Amman, Jordan. 2008–2012

Grants & Awards

- NSF National AI Research Resource (NAIRR) Pilot Grant 2025
  “Consensus-Seeking Autonomous Agents in Complex Cyber-Physical Transportation Systems” (Award No. NAIRR250132).
  National Science Foundation, USA. *Role: Co-Investigator.*

Fellowships & Scholarships

- University of Nebraska-Lincoln College of Engineering Professional Development Fellowship 2025, 2026
- Patrick T. McCoy Fund for Engineering Excellence 2025, 2026
- MATC & NTC Travel Fund Scholarship 2024, 2025, 2026

Experience

Graduate Research & Teaching Assistant

Jan 2023 – Present

University of Nebraska-Lincoln, Lincoln, NE, USA

Member of Dr. Wissam Kontar's research team at the Civil & Environmental Engineering Department and Nebraska Transportation Center. Research focuses on traffic agent heterogeneity and automated vehicle behavior control.

Teaching Assistantships:

- CIVE 202 – Civil Engineering Analysis II (programming in civil engineering)
- CIVE 361 – Principles of Transportation Engineering

Senior Transportation Engineer

Jun 2013 – Jan 2023

TrafQuest, Dubai, United Arab Emirates

Roles encompassed travel demand forecasting, traffic operations, multi-modal transport evaluation, microsimulation, safety analysis, geometric design, and technical reporting. Contributed to **100+ projects** across UAE and MENA, serving as Project Manager for the majority.

Project Category	Total	Engineer	Manager
Transportation Master Plan	20	12	8
Traffic Impact Study	64	24	40
Transportation Integration Study	11	1	10
Strategic Freight Transport Study	7	7	0
Parking Study	3	1	2
Traffic Management Plan	3	1	2
Evacuation Plan	1	0	1
Total	**109**	**46**	**63**

Graduate Research & Teaching Assistant

2011 – 2013

Jordan University of Science and Technology, Irbid, Jordan

Research in traffic operations, highway materials, and waste management: data collection, laboratory experiments, statistical analysis, and technical writing. Teaching: Surveying Lab, Highway Materials Lab, and Engineering Statics.

Mentorship

Undergraduate Research Mentorship

Mawunyo Tchamdja Jun – Aug 2025
NSF Research Experiences for Undergraduates (REU) Program
University of Nebraska-Lincoln, Lincoln, NE, USA.
Supervised undergraduate research focusing on autonomous electric mobility systems in Rural Nebraska as part of the program.

Service

Peer Review

Transportmetrica A: Transport Science (TR-A) ×2

KSCE Journal of Civil Engineering ×1
Bridging Transportation Researchers (BTR) Conference — Nos. 5, 6, 7, 8 ×4

Skills

Areas of Expertise

Automated Driving · Traffic Microsimulation · Reinforcement Learning · Machine Learning · Statistical Analysis · Optimization · Traffic Safety Analysis · Crash Analysis · Behavior Heterogeneity · Transport Modeling · Demand Forecast · Traffic Analysis · Systems Accessibility · Soft Mobility · Transport Integration · Transport Planning · Geometric Design

Software & Programming

Languages

Python · Julia · R · SPSS · SAS

Simulation & Planning

SUMO · VISSIM · VISUM · Synchro · SIDRA · HCS

GIS

ArcGIS · QGIS · GeoDa

Other

MS Office · LaTeX · HTML · Photoshop · Lightroom

References

Name	Role	Organisation	E-mail
Wissam Kontar	Assistant Professor	University of Nebraska-Lincoln	wkontar2@nebraska.edu
Jason Hawkins	Assistant Professor	University of Calgary	jfhawkin@ucalgary.ca
Ghazi Al-Khateeb	Professor / Chairman	University of Sharjah	galkhateeb@sharjah.ac.ae
Mohammad Braiwish	Managing Director	TrafQuest	m.younis@trafquest.com